Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2017

Centre Number

Candidate Number

Digital Technology

Assessment Unit AS 2

assessing

Fundamentals of Digital Technology

[CODE]

SPECIMEN PAPER

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
Write your answers in the spaces provided in this question paper.
Answer **all five** questions.
You are not permitted to use any calculating aid in this paper.

INFORMATION FOR CANDIDATES

The total mark for this paper is 100.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question of part question.
Quality of written communication will be assessed in questions **3(e)** and **5(d)**.

For Examiner's use only	
Question Number	**Marks**
1	
2	
3	
4	
5	
Total Marks	

Version 2: 6 March 2017

Examiner Only	
Marks	Re-mark

1 **(a)** What is meant by each of the following terms?

A kilobyte

_____

_____ [1]

A megabyte

_____

_____ [1]

In the two's complement system, positive and negative numbers can be represented in binary.

(b) Using the two's complement system, convert each of the following binary values to decimal, showing all your work.

00001111_2

[2]

11110110_2

[2]

(c) Show how the decimal number **− 44** can be represented as a two's complement binary number using 8 bits.

[4]

ASCII and Unicode are both used to represent characters.

(d) Describe how the bits in a byte are used in ASCII.

[5]

Examiner Only	
Marks	Re-mark

(e) Discuss how Unicode differs from ASCII.

Examiner Only	
Marks	Re-mark

[3]

2 Consider the following extract from a table in a stock control application.

ProductID	CurrentLevel	ReorderLevel
45667	343	400

(a) With the aid of examples, explain each of the following terms:

Data

[2]

Information

[2]

Knowledge

[2]

(b) Identify **two** factors which can affect the quality of information.

1

2

[2]

Examiner Only

Marks	Re-mark

In the ProductID **45667**, the **7** is a modulus 11 check digit.

(c) Show how this check digit is calculated.

[6]

Examiner Only	
Marks	Re-mark

(d) When completing an online application form, users must input their date of birth.

Discuss why both data validation and data verification should be performed in this case.

[6]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

3 **(a)** The internal components of a computer include a clock. Describe the purpose of the clock.

[3]

(b) Explain what happens during the fetch-execute cycle.

[3]

(c) Describe the main features of a WIMP user interface.

[4]

(d) Describe how a touch screen can be used to input text into a smart phone.

Examiner Only	
Marks	Re-mark

[4]

A photograph can be stored as a bitmap, JPEG or GIFF.

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(e) Evaluate all **three** of these formats for a photograph to be placed on a webpage.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark

4 The operating system manages all the operations of a computer.

(a) Identify **three** different tasks performed by a single user operating system.

1 _____

_____

2 _____

_____

3 _____

_____ [3]

Computer software includes utility programs.

(b) Describe the purpose of each of the following utilities.

Data compression software

_____

_____

_____

_____ [2]

Disk defragmenter software

_____

_____

_____

_____ [2]

Computer software also includes application software.

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Marks	Re-mark

(c) Describe each of the following types of software.

Generic software

_____

_____

_____

_____ [2]

Special purpose software

_____

_____

_____

_____ [2]

Banks use both batch processing and real-time processing.

(d) Describe how batch processing is used to produce monthly bank statements.

_____

_____

_____

_____

_____

_____

_____

_____ [3]

(e) Justify the use of real-time processing for customer transactions at Automated Teller Machines (ATMs).

Examiner Only	
Marks	Re-mark

[4]

5 **(a)** Explain each of the following web technology terms:

Examiner Only	
Marks	Re-mark

The WWW

[2]

The Internet

[2]

URL

[2]

(b) Explain how an intranet differs from the Internet.

[2]

(c) A page on a travel agent's web page contains the following text.

Examiner Only	
Marks	Re-mark

> Holiday types:
>
> - Flights only
> - Flights and hotel
> - Cruises
>
> Click here to continue to CCEAtravel.com

Produce the HTML code that will display this section of the web page.

_____

_____

_____

_____

_____

_____

_____

_____ [4]

(d) When paying for goods online, the user enters their personal and payment details and then clicks the Proceed icon. A short while later their screen shows the message 'Thank you for your order'.

By referring to this example, distinguish between client-side and server-side processing.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark